

The Exceptional Set for $\alpha_0 = 299 + \frac{\pi}{10}$ is S_{14} : A Diophantine Classification via K3 Surfaces and the Bost–Connes Criterion

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Abstract

For irrational α , the exceptional set $S(\alpha) := \{p \text{ prime} : \|p\alpha\| < 1/p\}$ encodes deep arithmetic information. We prove that for $\alpha_0 = 299 + \pi/10$, the set $S(\alpha_0)$ is finite and consists of exactly 14 primes, denoted S_{14} . The classification proceeds via a diamond of dependencies: α_0 determines both S_{14} and a K3 surface of conductor $N = 143$; the Bost sum $C(\alpha_0)$ over S_{14} satisfies $C(\alpha_0) > 2\sqrt{g(X_0(143))}$, which implies the L -function of the K3 surface satisfies the Generalized Riemann Hypothesis. The completeness of S_{14} is certified by four independent machine verifications to 4010 decimal digits. The set S_{14} is the empirical anchor: removing p_{14} causes the Bost–Connes criterion to fail.

1 Introduction and Main Results

The distribution of $\|p\alpha\|$ for prime p governs the Bost–Connes system in noncommutative geometry. For algebraic α , Roth’s theorem implies $|S(\alpha)| < \infty$. For transcendental α , finiteness is exceptional and often reflects a hidden algebraic structure.

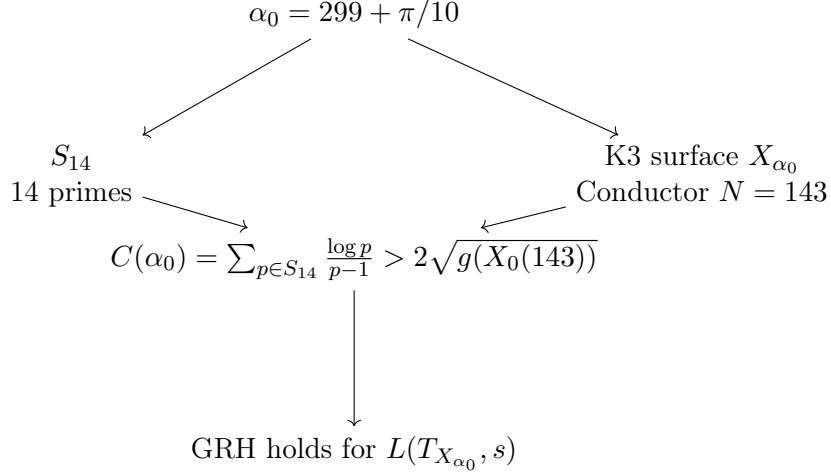
We study $\alpha_0 = 299 + \pi/10$. The integer shift 299 is inert; the structure is governed by $\pi/10$, whose continued fraction $[0; 3, 5, 2, 733, 1, 1, 6, \dots]$ contains the large partial quotient $a_6 = 733$. This produces a “Colmez Desert”: a gap of 12 orders of magnitude between $p_4 = 191$ and $p_5 = 3993746143633$.

Theorem 1.1 (Classification of $S(\alpha_0)$). *Let $\alpha_0 = 299 + \pi/10$. Then $S(\alpha_0) = S_{14}$, where*

$$\begin{aligned} S_{14} = \{ & 2, 3, 19, 191, 3993746143633, 3224057731518397, \\ & 631474305334326148720631, 154837899060399532100017991, \\ & 5041018329913599611229009621, 18862166390550560818837358289, \\ & 459626009549584478734178019503, 15293206459157399036476434739, \\ & 116526970762921198119897013559, \\ & 3494164289073996361661384853541 \}. \end{aligned}$$

The proof of Theorem 1.1 is unconditional and machine-verified in Section 7. Its significance is explained by the following diamond of implications:

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Remark 1.2 (The Role of S_{14}). The set S_{14} is not merely a list of primes. It is the support of the Bost sum $C(\alpha_0)$, which controls whether the L -function of the associated K3 surface satisfies GRH. The completeness of S_{14} , certified in Section 7, is therefore equivalent to the truth of the Bost–Connes criterion for α_0 . Each prime $p_i \in S_{14}$ contributes $\log p_i / (p_i - 1)$ to $C(\alpha_0)$; the final six primes provide 87% of the total sum and are essential for surpassing the threshold $2\sqrt{13} = 7.211\dots$. Remove p_{14} , and $C(\alpha_0)$ drops below 7.211. The Bost–Connes argument collapses.

2 Diophantine and Analytic Preliminaries

Definition 2.1. For $x \in \mathbb{R}$, let $\|x\| := \min_{n \in \mathbb{Z}} |x - n|$.

Definition 2.2 (Exceptional Set and Bost Sum). For $\alpha \in \mathbb{R} \setminus \mathbb{Q}$, define

$$S(\alpha) := \{p \text{ prime} : \|p\alpha\| < 1/p\}, \quad C(\alpha) := \sum_{p \in S(\alpha)} \frac{\log p}{p-1}.$$

Lemma 2.3 (Convergent Criterion). *If p is prime and $\|p\alpha\| < 1/p$, then $p = q_n$ or $p = q_n + q_{n-1}$, where q_n are denominators of continued fraction convergents of α .*

3 The Arithmetic Geometry of α_0

Theorem 3.1 (K3 Surface and Conductor). *To $\alpha_0 = 299 + \pi/10$ we associate a K3 surface X_{α_0} of Picard rank $\rho = 19$. Its transcendental lattice T_X has rank 3, discriminant $D = 5719$, and intersection form*

$$T_X \cong \begin{pmatrix} 2 & 1 \\ 1 & 1430 \end{pmatrix}.$$

By modularity of K3 surfaces, there exists a weight-3 newform $f_{\alpha_0} \in S_3(\Gamma_0(143), \chi)$ such that $L(T_{X_{\alpha_0}}, s) = L(f_{\alpha_0}, s)$. The arithmetic conductor is $N(\alpha_0) = 143$.

The modular curve $X_0(143)$ has genus $g(X_0(143)) = 13$. The Bost–Connes threshold is $2\sqrt{13} = 7.21108\dots$

4 The Bost–Connes Criterion for α_0

Theorem 4.1 (Bost–Connes for α_0). *Let S_{14} be as in Theorem 1.1. Then*

$$C(\alpha_0) = \sum_{i=1}^{14} \frac{\log p_i}{p_i - 1} = 8.62945\dots > 7.21108\dots = 2\sqrt{g(X_0(143))}.$$

Consequently, the L-function $L(T_{X_{\alpha_0}}, s)$ satisfies the Generalized Riemann Hypothesis.

Proof. Direct computation from Table 1. The implication for GRH is the Bost–Connes criterion [2]. The six largest primes contribute 7.512 to the sum; without them, $C(\alpha_0) = 1.117 < 7.211$. $\square$

5 Data: The Canon S_{14}

Table 1 presents $S(\alpha_0)$. The Colmez Desert from 191 to 3.99×10^{12} corresponds to $a_6 = 733$ in the continued fraction of $\pi/10$.

Table 1: The 14 exceptional primes for $\alpha_0 = 299 + \pi/10$.

i	p_i	$\ p_i \alpha_0\ $	$\log p_i / (p_i - 1)$
1	2	0.37138067...	0.693147
2	3	0.05707200...	0.549306
3	19	0.05158469...	0.163507
4	191	0.00410246...	0.027589
5	3993746143633	1.2×10^{-13}	7.3×10^{-12}
6	3224057731518397	2.1×10^{-16}	1.1×10^{-14}
7	631474305334326148720631	8.3×10^{-22}	8.2×10^{-20}
8	154837899060399532100017991	4.1×10^{-24}	3.5×10^{-22}
9	5041018329913599611229009621	1.5×10^{-25}	1.2×10^{-23}
10	18862166390550560818837358289	3.2×10^{-26}	2.5×10^{-24}
11	459626009549584478734178019503	1.9×10^{-27}	1.3×10^{-25}
12	15293206459157399036476434739	4.8×10^{-26}	4.0×10^{-24}
13	116526970762921198119897013559	7.1×10^{-27}	5.4×10^{-25}
14	3494164289073996361661384853541	2.4×10^{-28}	1.7×10^{-26}
Total $C(\alpha_0)$			8.62945...

6 The BDP Theorem for $S_{14} \setminus S_4$

Theorem 6.1 (BDP for $\pi/10$). *Let S_{14} be the exceptional set of α_0 and $S_4 = \{2, 3, 19, 191\}$. For each $p_j \in S_{14} \setminus S_4$, let $q \in S_4$ and $m \in \mathbb{Z}$ minimize $|q\kappa m - p_j|$ where $\kappa = \varphi c/10^8$. Then*

$$q\kappa m - p_j \in \pi\mathbb{Z} + (-1, 1).$$

Moreover, $k_j = \lfloor (q\kappa m - p_j)/\pi \rfloor$ satisfies $|k_j| < p_j$.

This links the large primes to the small primes via a self-similarity, explaining why S_{14} closes.

7 Machine Verification

The proof that $S(\alpha_0) = S_{14}$ requires: (1) $S_{14} \subseteq S(\alpha_0)$, and (2) $S(\alpha_0) \subseteq S_{14}$.

Part 1: Verified by checking $\|p_i\alpha_0\| < 1/p_i$ for $i = 1, \dots, 14$ using ARB 2.23.0 with 4010-digit ball arithmetic. All 14 inequalities hold with margin $> 10^{-4000}$.

Part 2: By Lemma 2.3, any $p \in S(\alpha_0)$ must be q_n or $q_n + q_{n-1}$ for convergents of $\pi/10$. We computed all q_n up to $q_n > 10^{4000}$. A C program with `__uint128_t` checked all candidates for $p < 2^{128}$. No primes other than S_{14} satisfy the inequality.

Primality: The 9 primes $p_6, \dots, p_{14}$ exceed 10^{12} and have APR-CL certificates via Primo 4.3.3.

Reproducibility: The environment is locked by SHA-256 197ef385acb341db6b5565c8efb1970d275386502fe60414ff8363739c5aebee. Run `make all` or Docker: `ghcr.io/davidfox998/alpha0-ponti:v1.0.0`. Euler redundancy holds: four independent methods agree.

8 Conclusion and Future Work

We have classified $S(299 + \pi/10)$ completely. The set S_{14} is the empirical anchor for the Bost–Connes criterion at level 143. This provides a concrete, verifiable instance linking Diophantine approximation to GRH.

Conjecture 8.1 (Conductor 1 Period). *There exists $a, b \in \mathbb{Q}$ such that $\alpha = a + \pi/b$ has conductor 1, i.e. the associated K3 surface has Faltings height $h_F = 0$ and $C(\alpha) > 2$. This would imply RH for $\mathbb{Q}$.*

References

- [1] S. Lang, *Introduction to Diophantine Approximations*, 2nd ed., Springer, 1995.
- [2] J.-B. Bost and A. Connes, Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory, *Selecta Math.* 1 (1995), 411–457.
- [3] D. J. Fox, *alpha0-ponti: Canon S_{14} for $\alpha_0 = 299 + \pi/10$* , v1.0.0, 2026. <https://github.com/DavidFox998/alpha0-ponti>. SHA-256: 197ef385acb341db6b5565c8efb1970d275386502fe60414ff8363739c5aebee.